

PROJECT DESCRIPTION	PROJECT NAME		TIT PCI-DB VER0.1-1.2T		TOLERANCE RULE	B	0.1% TOLERANCE
	DESIGNER		sey@titeng.co.kr			D	0.5% TOLERANCE
	DATE		2026. 02. 11			F	1.0% TOLERANCE
	FILE NAME		TITPCI-DB V01_260211.DSN			J	5.0% TOLERANCE
REVISION HISTORY	NO	VERSION	DATE	DESCRIPTION		K	10% TOLERANCE
	01	VER0.1	2026. 02. 11	NEW DESIGN		M	20% TOLERANCE
	02					Z	+80% ~ -20% TOLERANCE
	03						
	04						
	05						
	06						
PAGE LIST	D01. COVER D02. BLOCK DIAG1 S01. MCU S02. FLASH S03. SECURE_BOX S04. GPIO_NFC S05. UART_USB S06. LED S07. POWER S08. BACKUP_BAT				ARTWORK GUIDE	01. LAYOUT은 배치 검토 후 진행 02. 회로도 오류는 협의 후 결정 03. 모든 전원은 PATTERN 처리 후 내층 연결 04. BYPASS CAPACITOR는 용량이 작은 것을 IC에 가깝게 배치 05. 작업 완료 시 BOARD에 날짜와, 이름 명기 06. 20PIN 이상의 부품은 5PIN마다 선표시, 10PIN마다 번호 표기할 것 07. BOARD는 원칙적으로 COPPER 처리할 것 08. 각 TP와 CONNECTOR 이름은 회로도에 기록된 이름을 사용 09. 그 외의 사항은 협의 후 결정	
					PCB LAYER	FR-4 / 무연 HAL / 4-LAYER / 1.2T / GREEN	

CAUTION !!!

THIS SCHEMATIC IS THE SCHEMATIC OF SYSTEM ONLY FOR DEVELOPMENT IN THE TITENG. CO., LTD.
IF THIS SYSTEM IS CONNECTED WITH OTHER SYSTEM, THIS SYSTEM MA BE OUT OF ORDER.

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Title

D01. COVER

Size

Document Number

Rev

A3

TITENG-260324

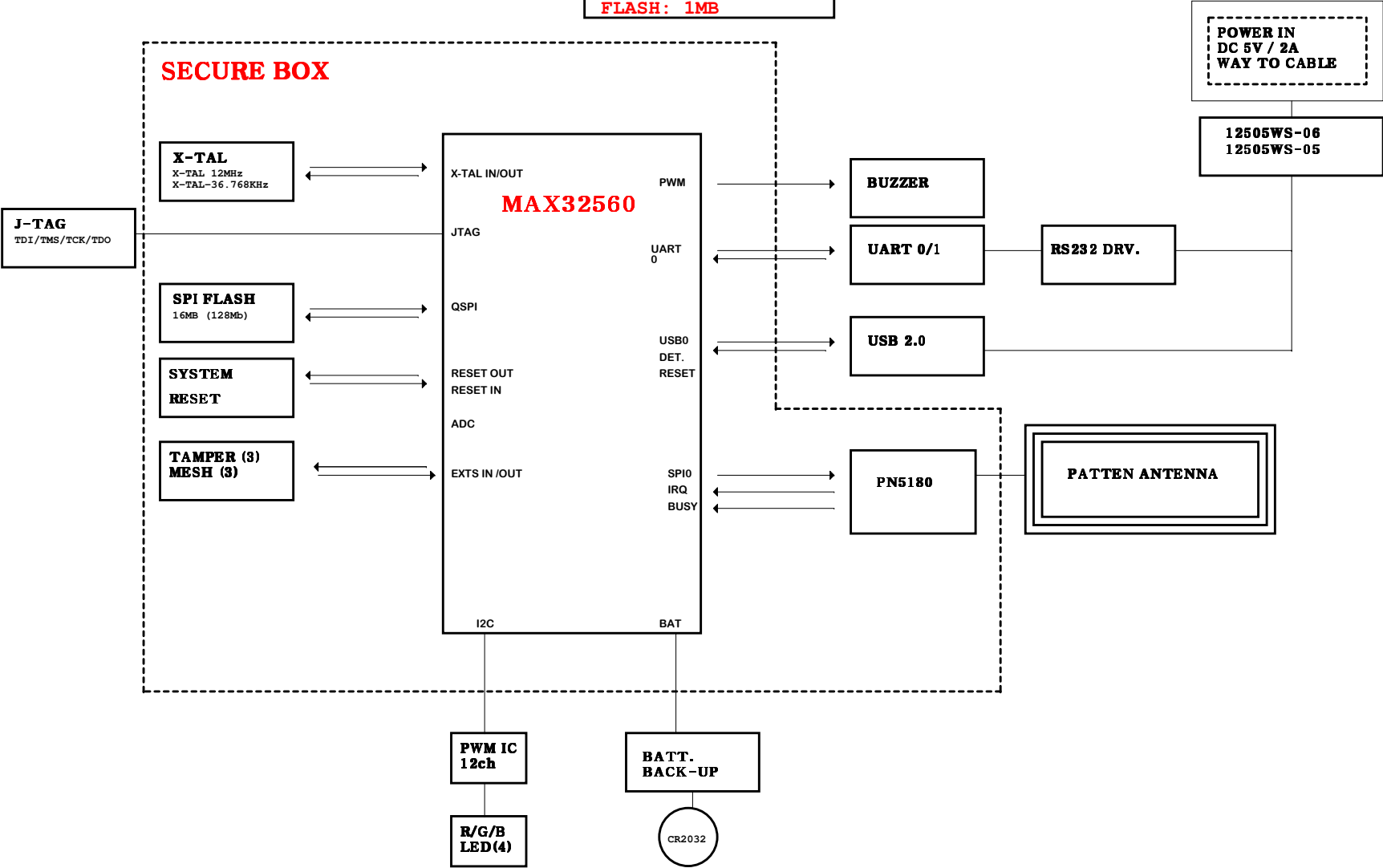
0.2

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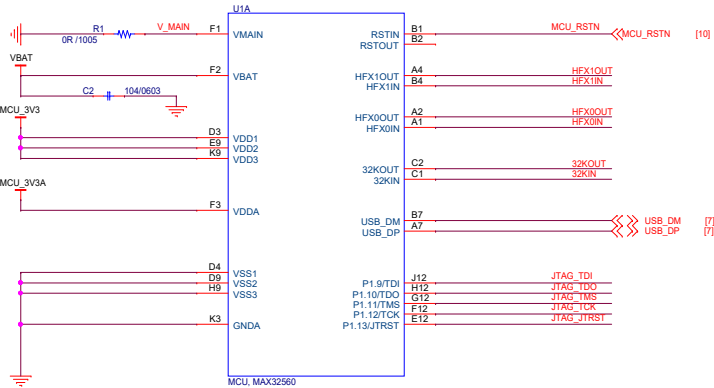
BLOCK DIAGRAM (MAX32560)

32-bit RISC CPU
Arm Cortex-M3
108.48MHz
SRAM: 384KB (2bank)
FLASH: 1MB

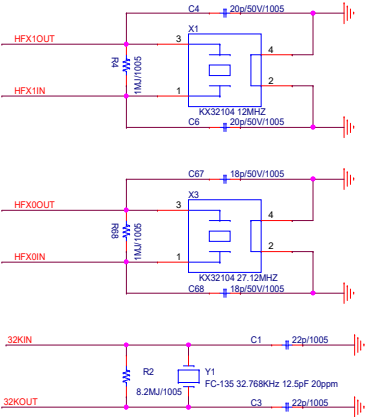


SECURE MCU(MAX32560)

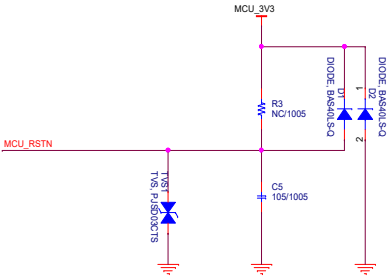
SYS PART



X-TAL(SYS/32K)

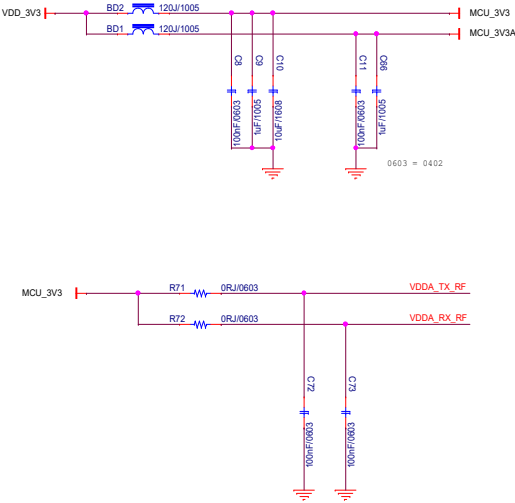
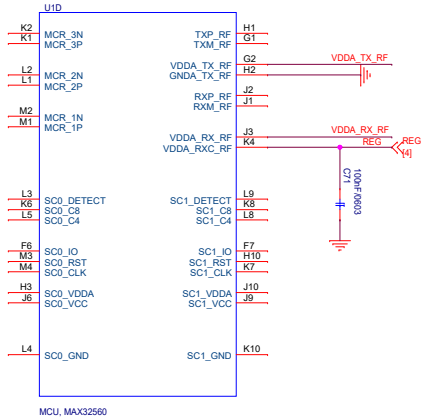


RESET

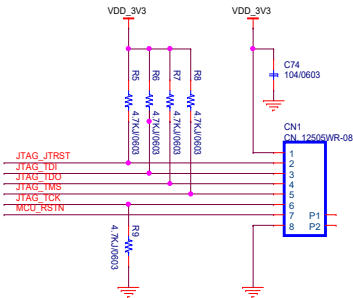


internal pullup resistor in the range of 50 kΩ-150 kΩ

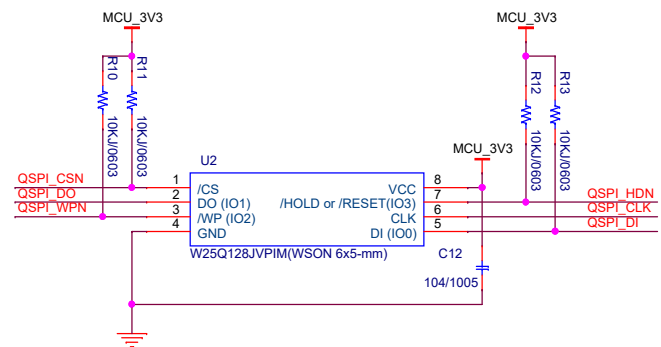
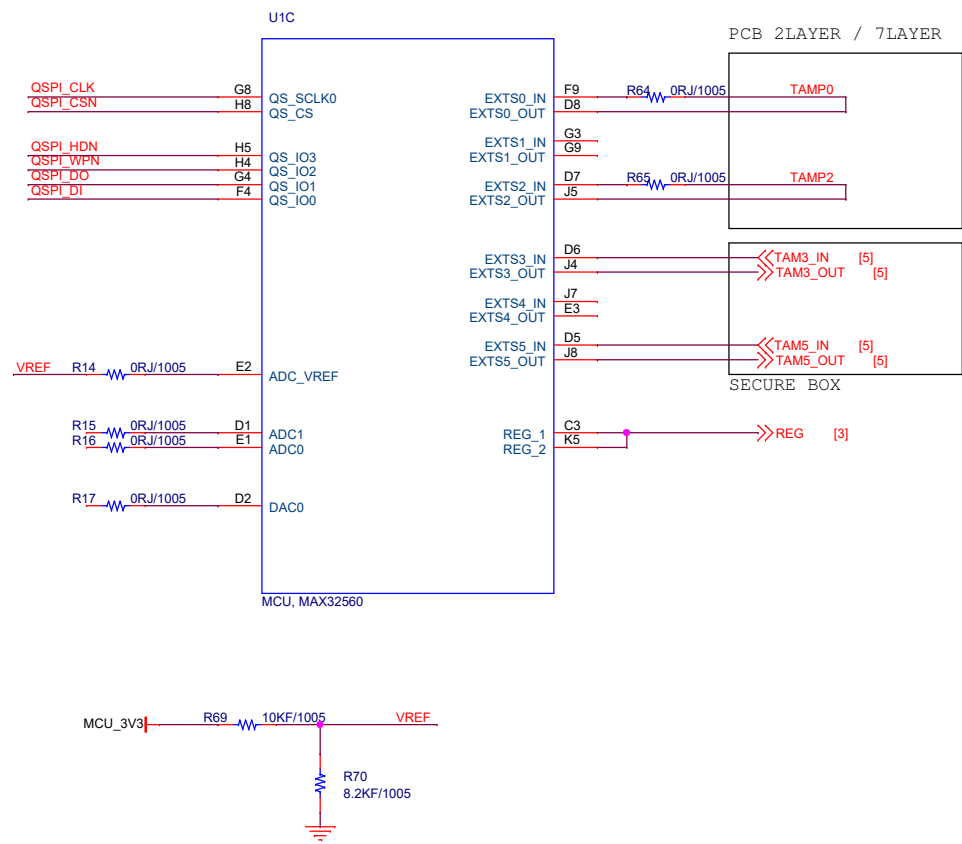
MSR/ IFM /SAM PART (NC)



ARM JTAG



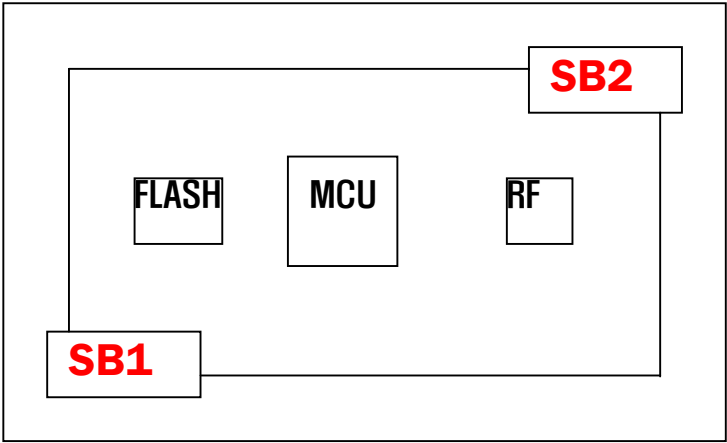
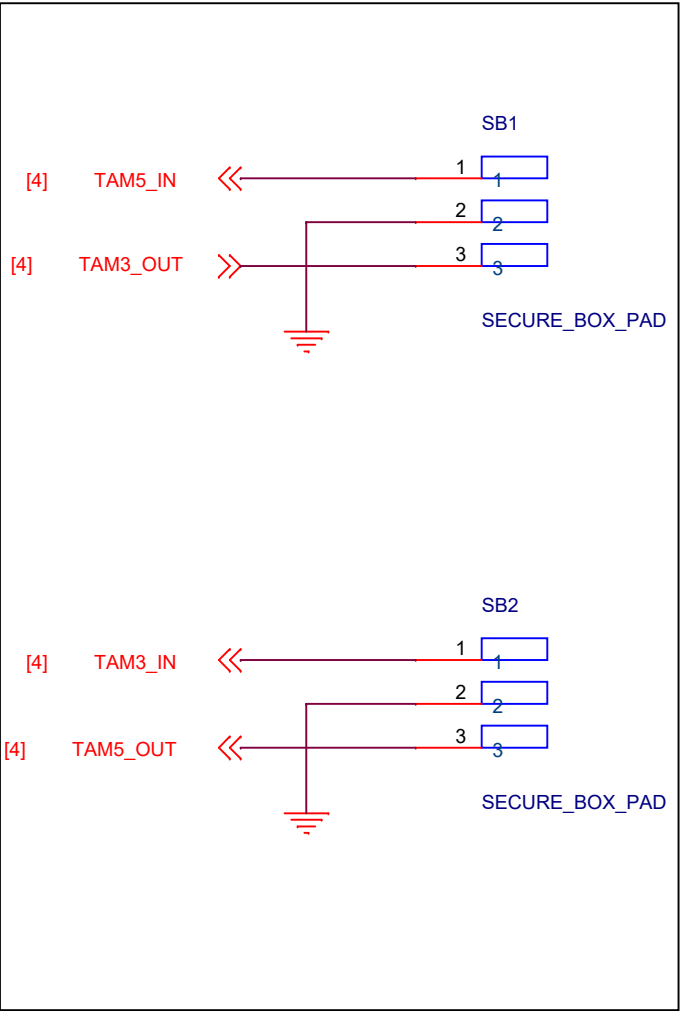
SPI FLASH / TAMPER



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SECURE BOX

TAMPER S/W 3, 5 PAIR



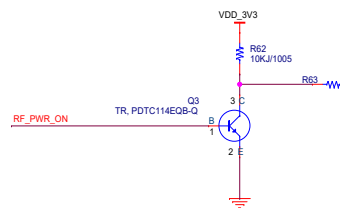
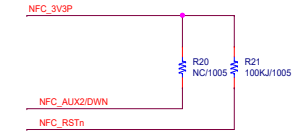
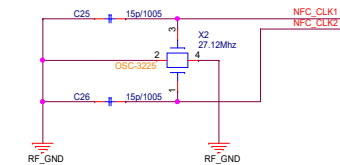
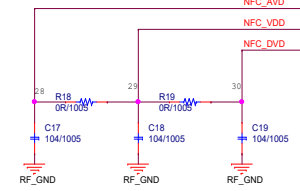
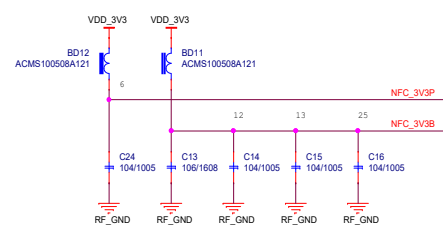
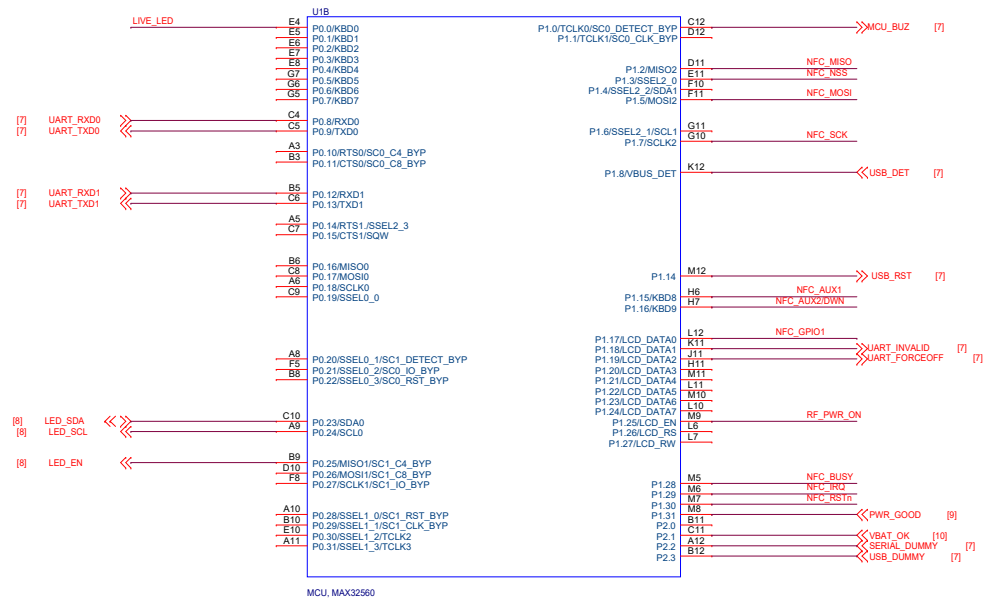
MESH PATTEN (TAMPER 0,1,2 PAIR)

PCB LAYER1	TOP MOUNT (VIA TENTING)
PCB LAYER2	MESH (W:0.2mm/ S:0.2mm)
PCB LAYER3	GND
PCB LAYER4	VCC
PCB LAYER5	TINNER PATTEN
PCB LAYER6	GND
PCB LAYER7	MESH (W:0.2mm/ S:0.2mm)
PCB LAYER8	BOTTOM MOUNT (VIA TENTING)

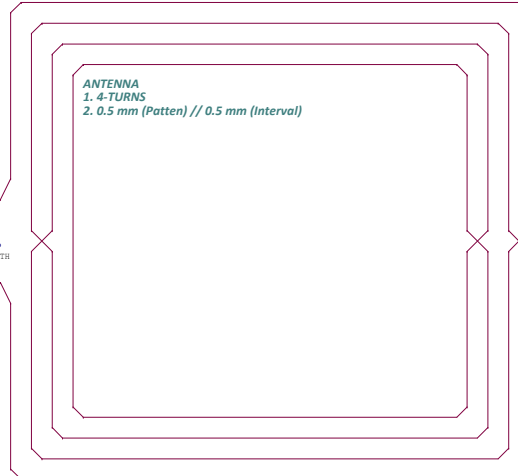
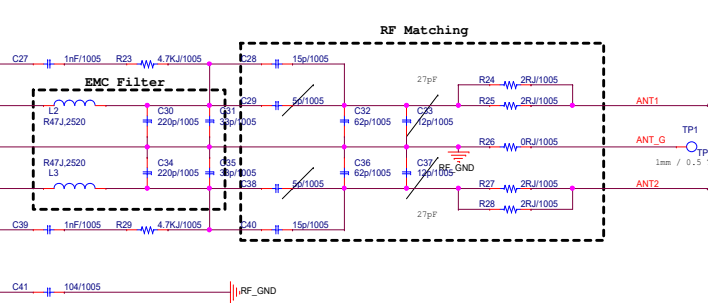
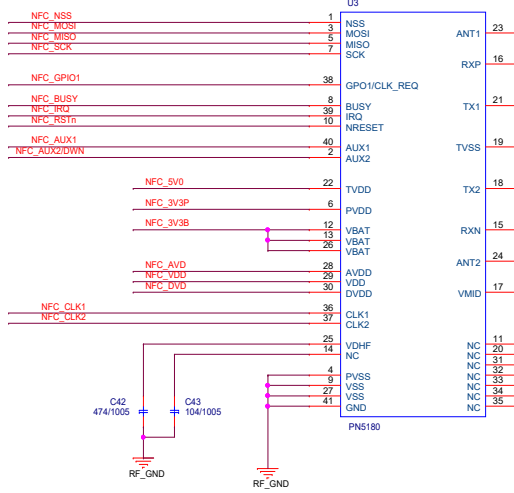
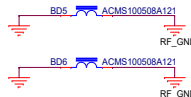
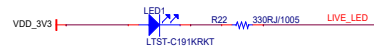
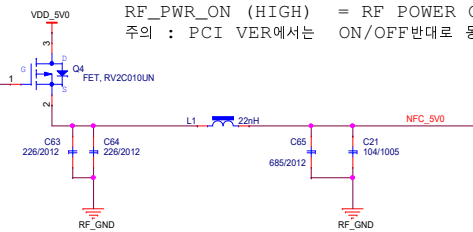
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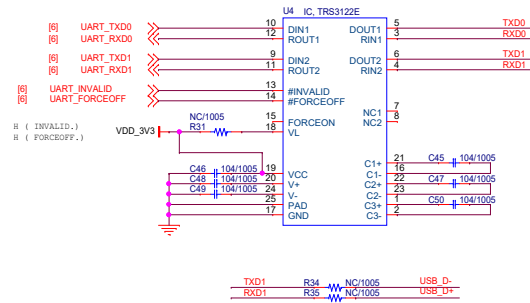
Title		
S03. SECURE_BOX		
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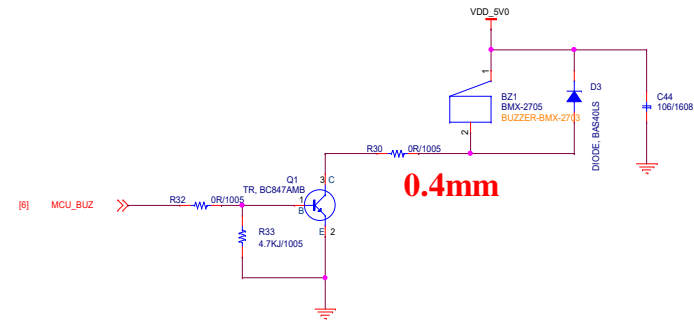
RF_PWR_ON (LOW) = RF POWER ON
 RF_PWR_ON (HIGH) = RF POWER OFF
 주의 : PCI VER에서는 ON/OFF반대로 동작함.



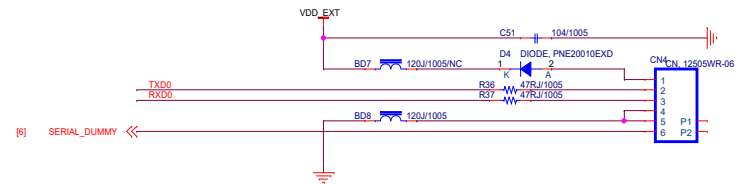
UART0/1 DEBUG



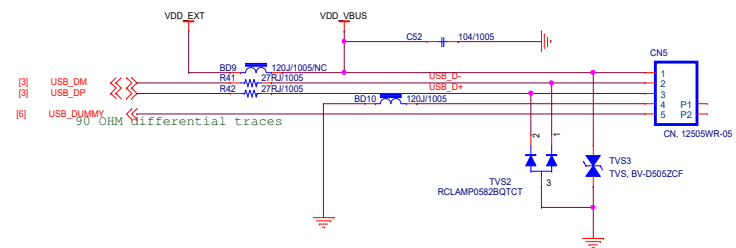
BUZZER



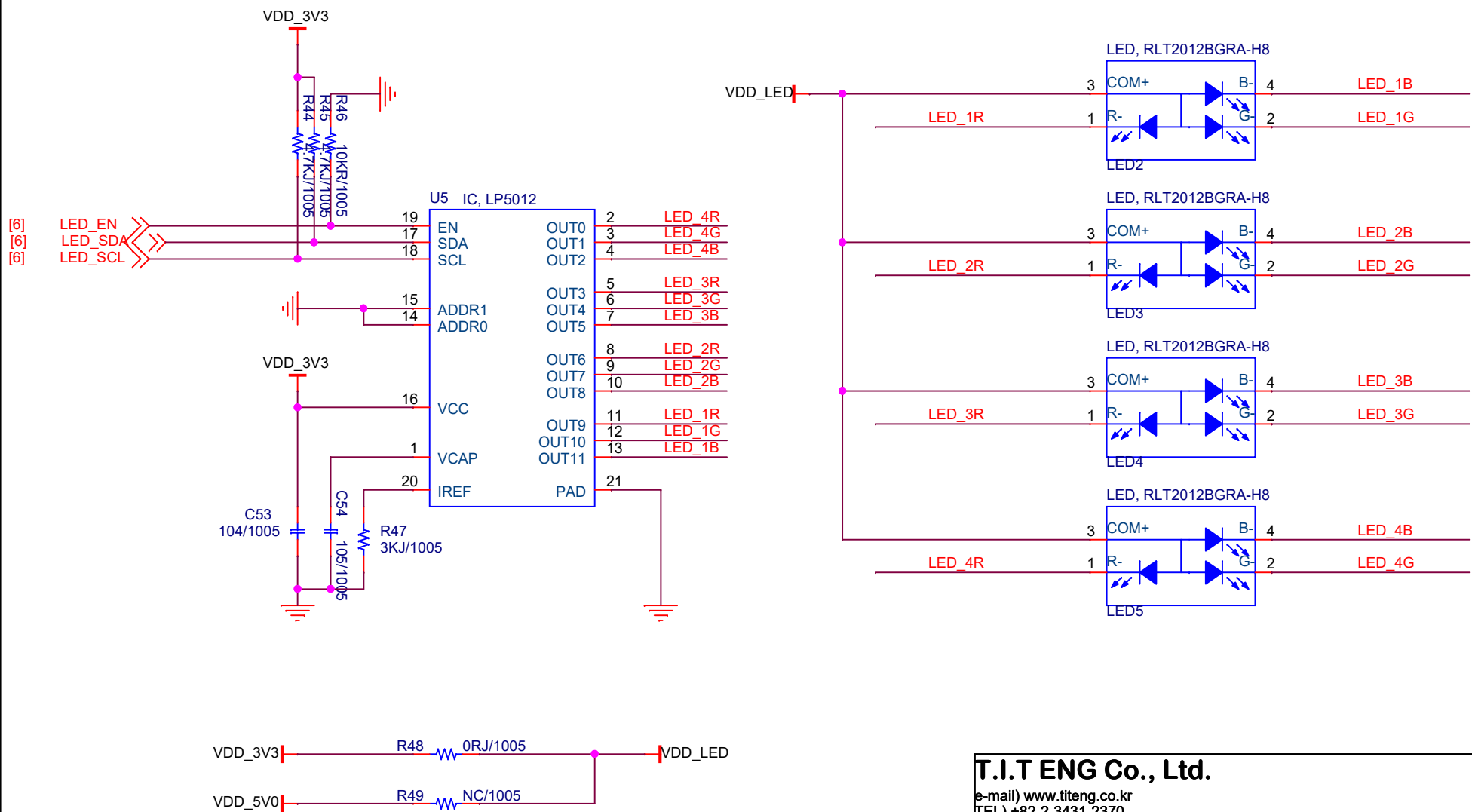
SERIAL PORT



USB PORT



R,G,B LED - 4CH PWM



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S06. LED

Size

Document Number
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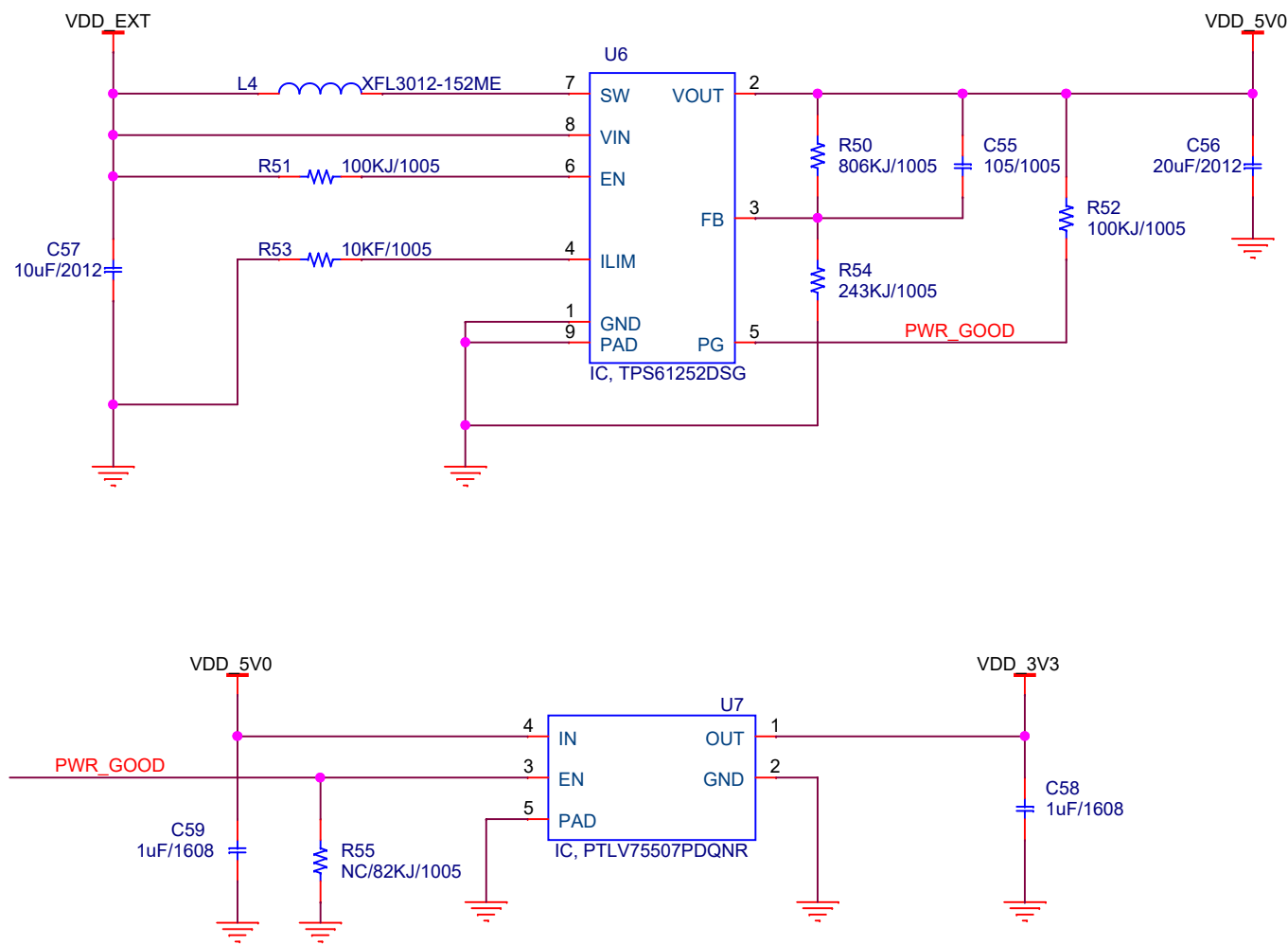
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POWER (5.0 / 3.3V)



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S07. POWER

Size

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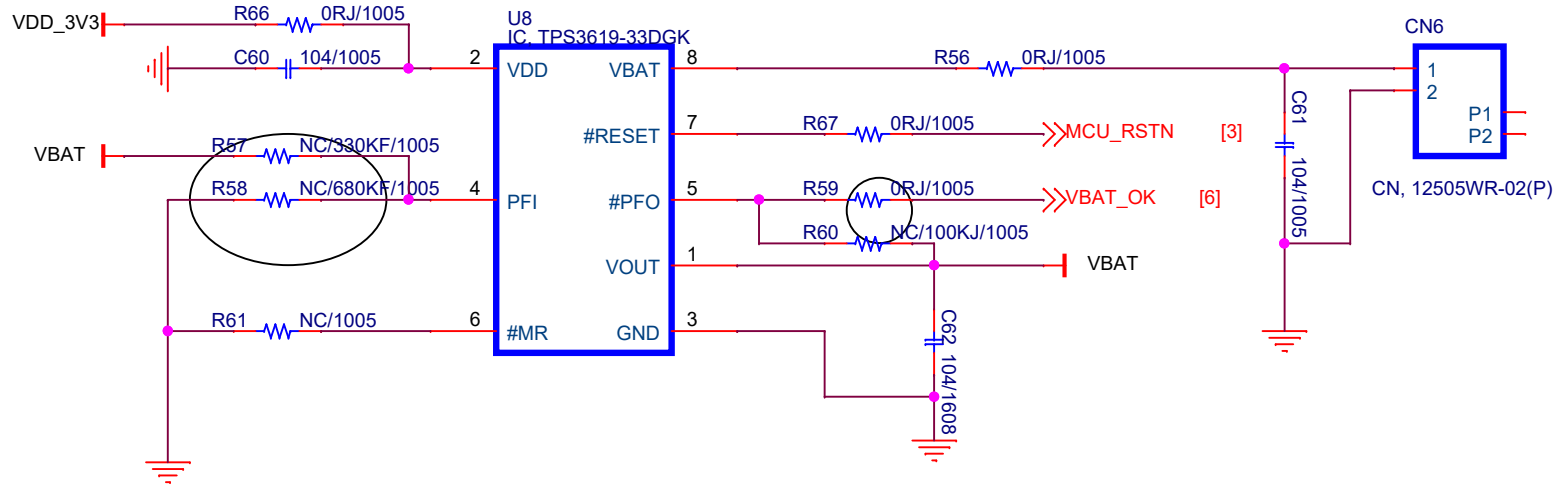
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BATTERY CONNECTOR (CR2032)

COIN CELL 3V / 210-240mAh / Li-MnO2



BATT. MONITOR 2.0V이하 일때 #PFO =

HIGH
 $VPFI(1.18V)=2.0 \times 0.68=1.36V$

$VPFI(1.18V)=3.0 \times 0.68=2.04V$

NONE-ACTIVATION MODE(BACKUP MODE)

IC 동작 전류: 3.9 μA @25°C

누설전류: 0.1 μA

BACKUP IC: 0.1 μA

총전류=3.9+0.1+0.1=4.1 μA

240/0.0041=58,536h / 6.68 years

*자가방전 및 온도 / PCB손실 등 약 5년~6년 예상

@85°C 적용시 4~5년예상.

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S08. BACKUP_BAT

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